Delineating Arterial/Venous Pulse Waveform at Forehead/Hand using Webcam

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Abstract
This study presents a non-contact method for estimating the arterial pulse rate using a standard webcam. The methodology involves detecting the subject's face, extracting the forehead region, and analysing the temporal variations in the green channel intensity. These periodic variations correspond to pulsatile blood flow, allowing us to estimate the heart rate. Signal processing techniques are applied to enhance the signal and identify peaks accurately. The results demonstrate the viability of webcam-based Pulse Rate (Heart Rate) for real-time pulse monitoring. The same approach is extended to hand regions, validating the method's versatility across multiple anatomical sites.

Keywords- Pulse Waveform, Webcam, Green Channel, Face Detection, Signal Processing, Heart Rate, Forehead, Hand, Remote-PPG.

I. Introduction

Traditional Pulse/heart rate monitoring relies on contact-based sensors like ECG or fingertip PPG sensors. Recent advancements in computer vision and signal processing have made it possible to detect remote pulses using video imaging. In this project, we explore a non-contact technique to estimate pulse rate by analysing colour intensity variations in skin regions captured by a webcam in real time, as well as a complete plot for the waveform to analyse it for different properties for different uses. Specifically, we extract the green channel from the forehead or hand, as it is most responsive to blood volume changes due to its absorption properties.

II. Methodology

A. System Overview

A standard webcam is used to record video of the subject continuously using the Opencv Python library. MediaPipe's face detection module is utilised to locate facial landmarks, focusing on the forehead region. The green channel intensity is recorded and processed to identify periodic peaks corresponding to heartbeats.

B. Face and Region Detection

Using MediaPipe's FaceDetection, we detect bounding boxes around faces. The forehead is isolated by slicing the face rectangle's top 20-30%. For hand-based detection (future implementation), similar logic will be adapted.

Figure 1 – Forehead Region detection from face using webcam, showing real-time heart rate after a few seconds.

Figure 2 – Putting the hand in the marked region to extract colour components.

C. Signal Acquisition and Processing

We extract the average green channel value per frame from the forehead region. These values are stored along with timestamps. A moving average filter smooths the signal, and the find_peaks function from scipy.signal is used to detect heartbeat-like peaks.

D. Heart Rate Estimation

The intervals between peaks are computed to calculate the average peak-to-peak time. The pulse rate in beats per minute (BPM) is then calculated using:

$$Heart\ Rate\ (BPM) = \frac{60}{Average\ Time\ Interval\ Between\ Beats\ (sec)}$$

III. Code Implementation

The implemented system utilises a webcam to capture real-time video frames and extract physiological signals non-invasively from predefined regions of interest (ROIs) — either the forehead (using face detection) or a fixed rectangular box positioned over the forearm. From each frame, the green channel intensity is averaged within the ROI, as it is highly responsive to blood perfusion. These values are recorded over time, smoothed using a moving average filter, and analysed for periodic peaks corresponding to heartbeats using scipy.signal.find_peaks. The time intervals between successive peaks are then used to calculate the heart rate in beats per minute (BPM). The detected pulse waveform is finally visualised using Plotly for signal analysis and validation.

Note: Due to IEEE restrictions, code will only be abbreviated or explained using simple language.

IV. Results

- Green channel values successfully captured from the forehead region.
- A smooth periodic waveform was observed after applying a moving average filter.

- Accurate peak detection enabled pulse estimation with results typically between 60–110 BPM.
- Visualisation using Plotly confirms the detection of valid PPG waveforms from webcam data.

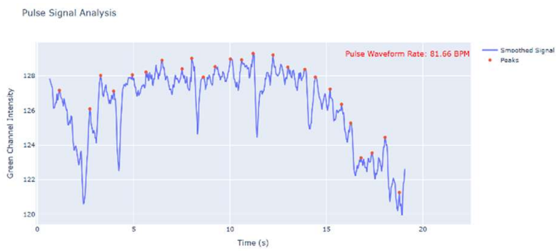

Figure 3 – This type of plot is expected to show the peaks and a heart rate value, which we are concluding as the pulse rate.

V. Discussion

This method leverages the photoplethysmographic principle by utilising light absorption differences caused by blood volume changes remotely using the webcam only. The green channel shows the highest sensitivity to pulsatile flow. Despite challenges like motion artefacts and lighting, reasonable accuracy is possible.

Future work includes:

- Enhancing robustness under different lighting conditions.
- Aiming for better accuracy and real value with standard PPG/ECG devices.
- Make it available for all the remote devices, including smartphones, for easy access with a simple interface application.

VI.Conclusion

This project demonstrates a simple, non-contact method for pulse rate detection using a regular webcam and basic signal processing. The system is low-cost, accessible, and can be a starting point for remote health monitoring solutions. The technique aims to apply this approach to the Forehead Region and the Hand Region for continuous detection.

VII. References

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